

1.1 The main hydrological characteristics and processes that operate in rivers and drainage basins

This chapter will explain:

- ★ the characteristics of rivers and drainage basins
- ★ how the drainage basin operates within the water cycle
- ★ the processes that operate in a drainage basin
- ★ the processes that operate within a river.

The characteristics of rivers and drainage basins

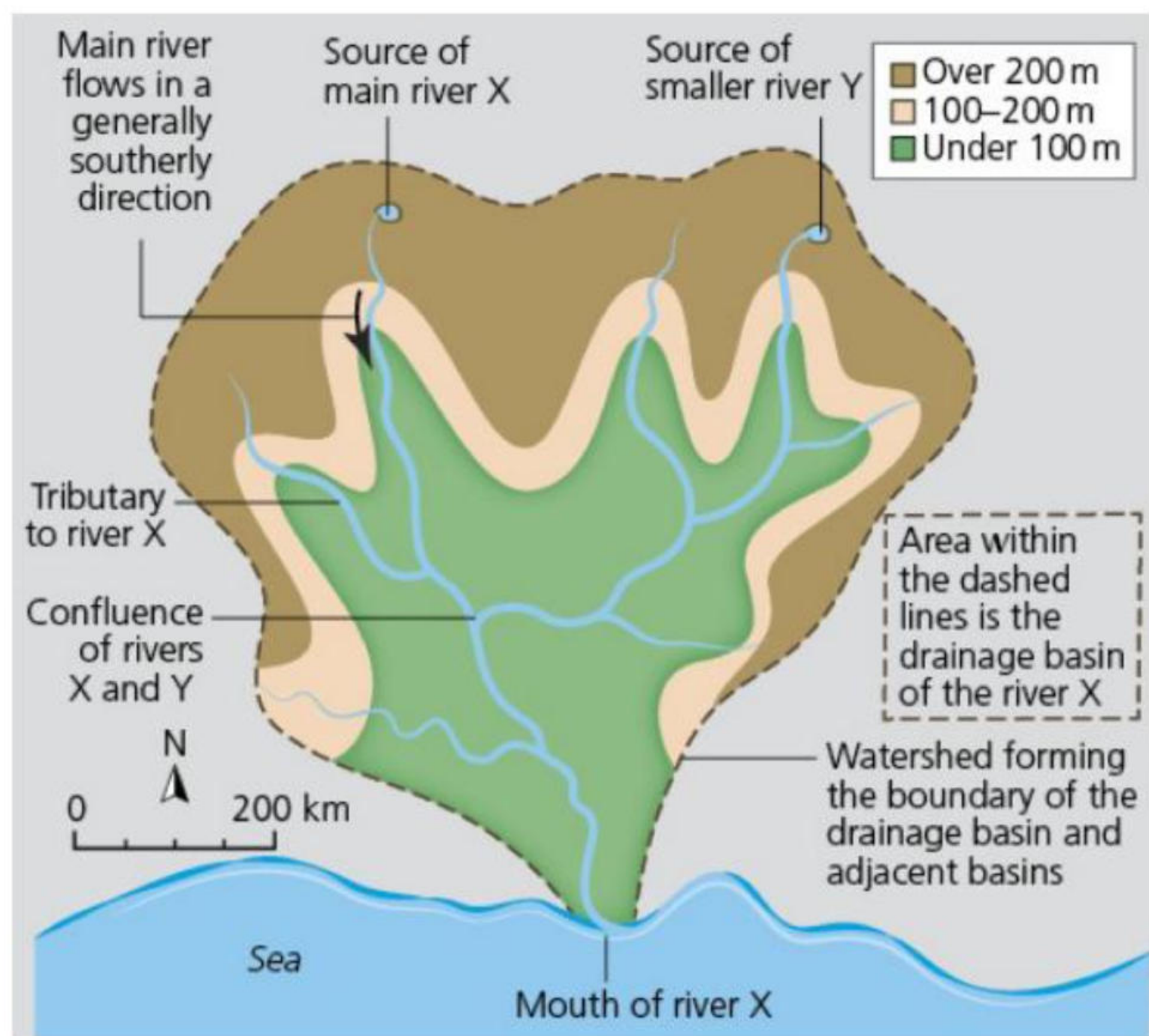
A **drainage basin** (or catchment area) is the area drained by a river and its tributaries. Some drainage basins are very small (such as less than 10 km²). The largest drainage basins are huge, however – the Mississippi River and its tributaries drain over one-third of the USA ([Table 1.1](#)).

▼ **Table 1.1** The drainage basins of some of the world’s major rivers

River	Continent	Length (km)	Area of drainage basin (km ²)	Average discharge (m ³ /sec)
Amazon	South America	6387	6,144,727	219,000
Nile	Africa	6690	3,254,555	5100
Mississippi/Missouri	North America	6270	3,202,230	16,200
Yangtze	Asia	6211	1,800,000	31,900

Drainage basins have a number of distinct features ([Figure 1.1](#)):

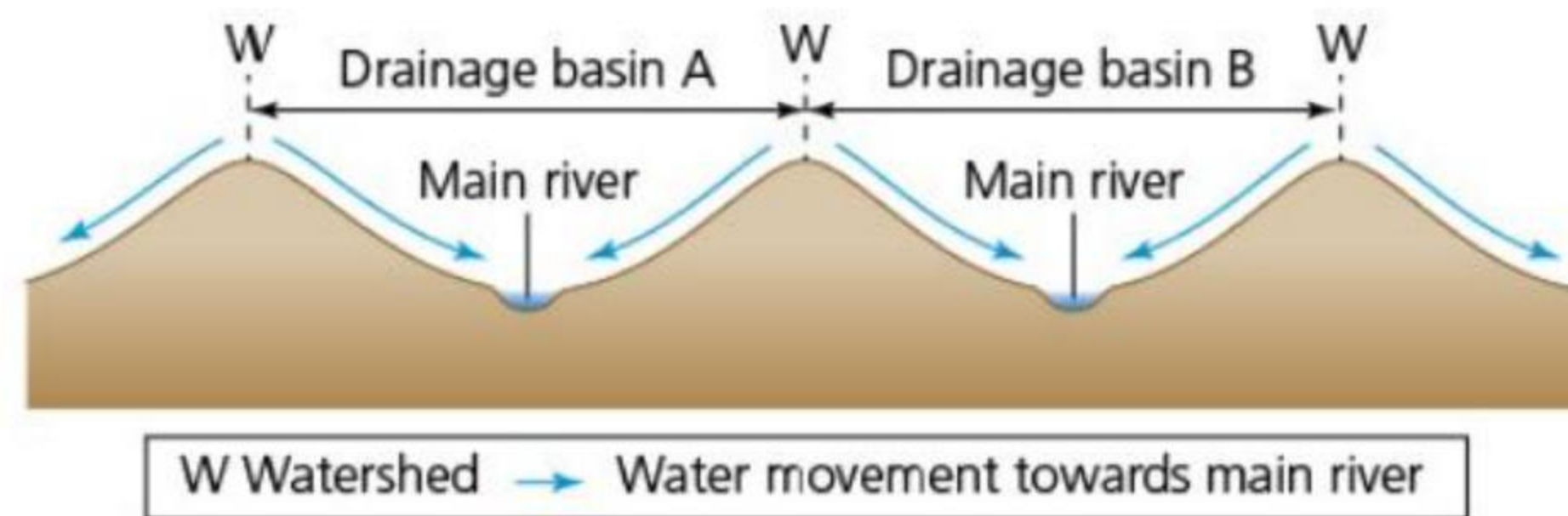
- » The boundary of a drainage basin is called the **watershed**.
- » The point where a river begins is known as its **source**.
- » A river reaches the sea at its **mouth**.
- » A **tributary** joins the main river at a **confluence**.
- » A main river and all its tributaries form a **channel network** or river system.



▲ **Figure 1.1** The features of a drainage basin

A watershed ([Figure 1.2](#)) is a ridge of high land that forms the boundary between one drainage basin and

other adjacent basins.



▲ **Figure 1.2** A cross-section showing drainage basins and watersheds

The source of a river

A river is a large, natural stream of flowing water. The place where a river begins may be:

- » an upland lake – the Mississippi River, the largest river in North America, begins as a stream from Lake Itasca in the US state of Missouri
- » a melting glacier ([Figure 1.3](#)) – the Gangotri Glacier in the Himalaya mountains is the source of the River Ganges in Asia



▲ **Figure 1.3** A meltwater stream emerging from the Fox Glacier in the Southern Alps, New Zealand

- » a spring in a boggy upland area, where the soil is so saturated that recognisable surface flow begins – the source of the Danube River is a spring in such an area of the Breg River, in the Black Forest in Germany
- » a spring at the foot of an escarpment at the boundary between permeable and impermeable rock ([Figure 1.4](#)) – there are many such springs at the foot of the North Downs and South Downs, in southeast England.

When small streams begin to flow, they act under gravity, following the fastest route downslope. As they take the lowest path in the local landscape, water is added to them from tributaries, **groundwater flow**, **throughflow** and **overland flow** (surface runoff).

Channel networks

Some main rivers have a large number of tributaries, meaning that no place in the drainage basin is very far from a river. Such an area is said to have a high **drainage density**. The Amazon River, for example, receives water from more than a thousand tributaries. Where a main river has few tributaries, the drainage density is low. Channel networks often form a distinct pattern, which is due to the structure of rocks in the drainage basin.



▲ **Figure 1.4** Water issuing from a spring at the boundary of permeable and impermeable rock, Malham Cove, Yorkshire, UK; this is the source of Malham Beck

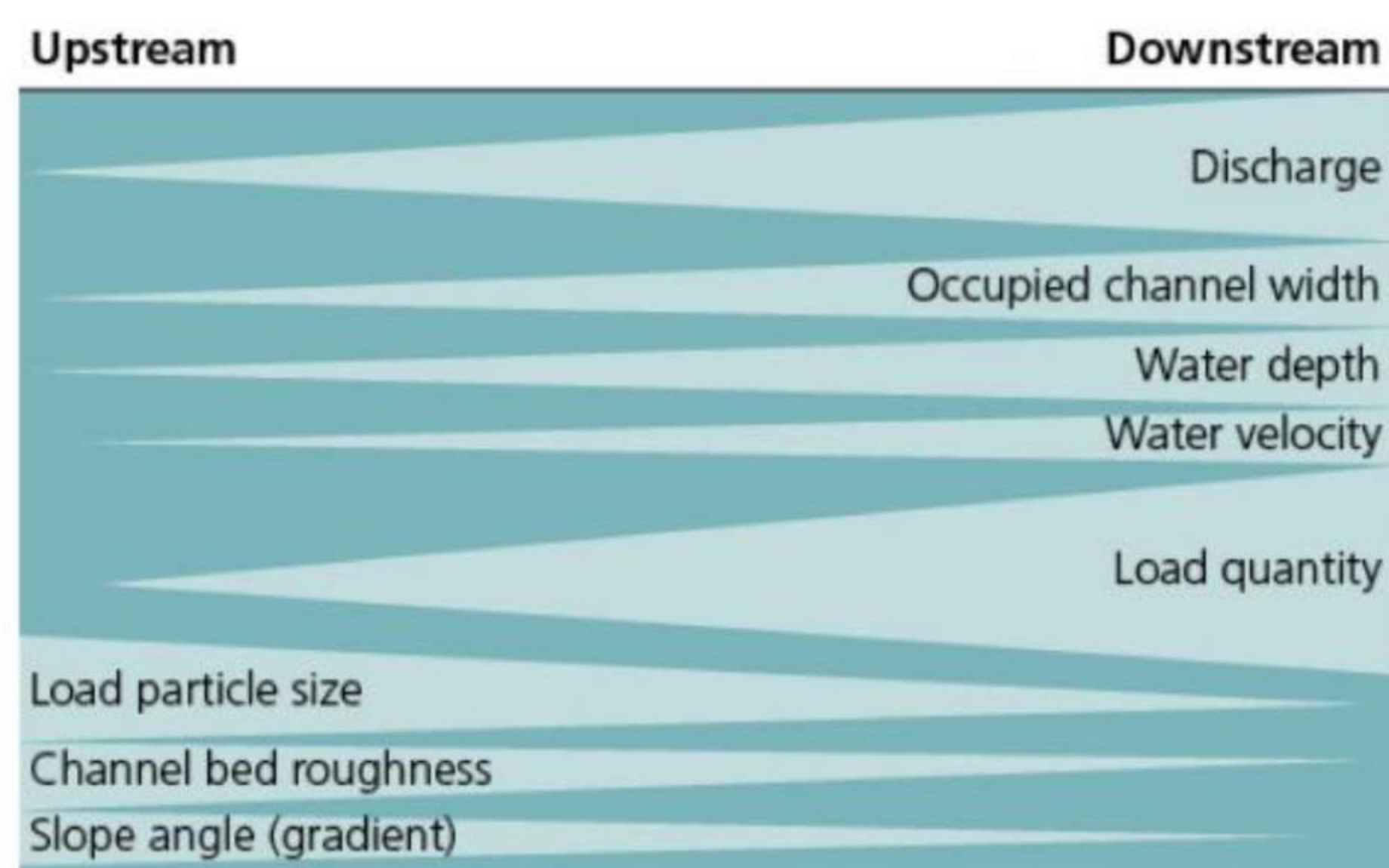
The mouth of a river

A river mouth occurs where a river empties into another body of water – a larger river, a lake or a sea or ocean. The great majority of rivers drain into a sea or ocean, but some drain into lakes that may be far from a coastline. For example, the River Volga, the longest river in Europe (at approximately 3685 km), flows into the inland Caspian Sea. Deltas sometimes form at the mouth of a river where the strength of tides and currents is insufficient to clear the large-scale sediment arriving from further upstream. The biggest delta in the world is the Ganges delta in Bangladesh and India.

The long profile

The **long profile** of a river is a longitudinal section of the course of the river, drawn along the river from source to mouth. It is expressed graphically as a curve, with the idealised form being a concave-upwards curve. [Figure 1.5](#) shows how rivers change along their course from upstream to downstream:

- » Discharge, width, depth, speed of flow/velocity and load quantity all increase.
- » Load particle size, channel bed roughness and the river's gradient all decrease.



▲ **Figure 1.5** Changes in a river from upstream to downstream

Channel shape

The efficiency of a stream's shape is measured by its **hydraulic radius** – that is, the cross-sectional area divided by the **wetted perimeter** ([Figure 1.6](#)). The wetted perimeter is the total length of the bed and the bank sides that are in contact with the water in the channel. The higher the ratio, the more efficient the stream and the smaller the frictional loss. The ideal form is semi-circular.

[Figure 1.6](#) shows two channels with the same cross-sectional area, but with different shapes and hydraulic radii:

- » Stream A: With a larger hydraulic radius, this stream has a smaller amount of water in contact with the

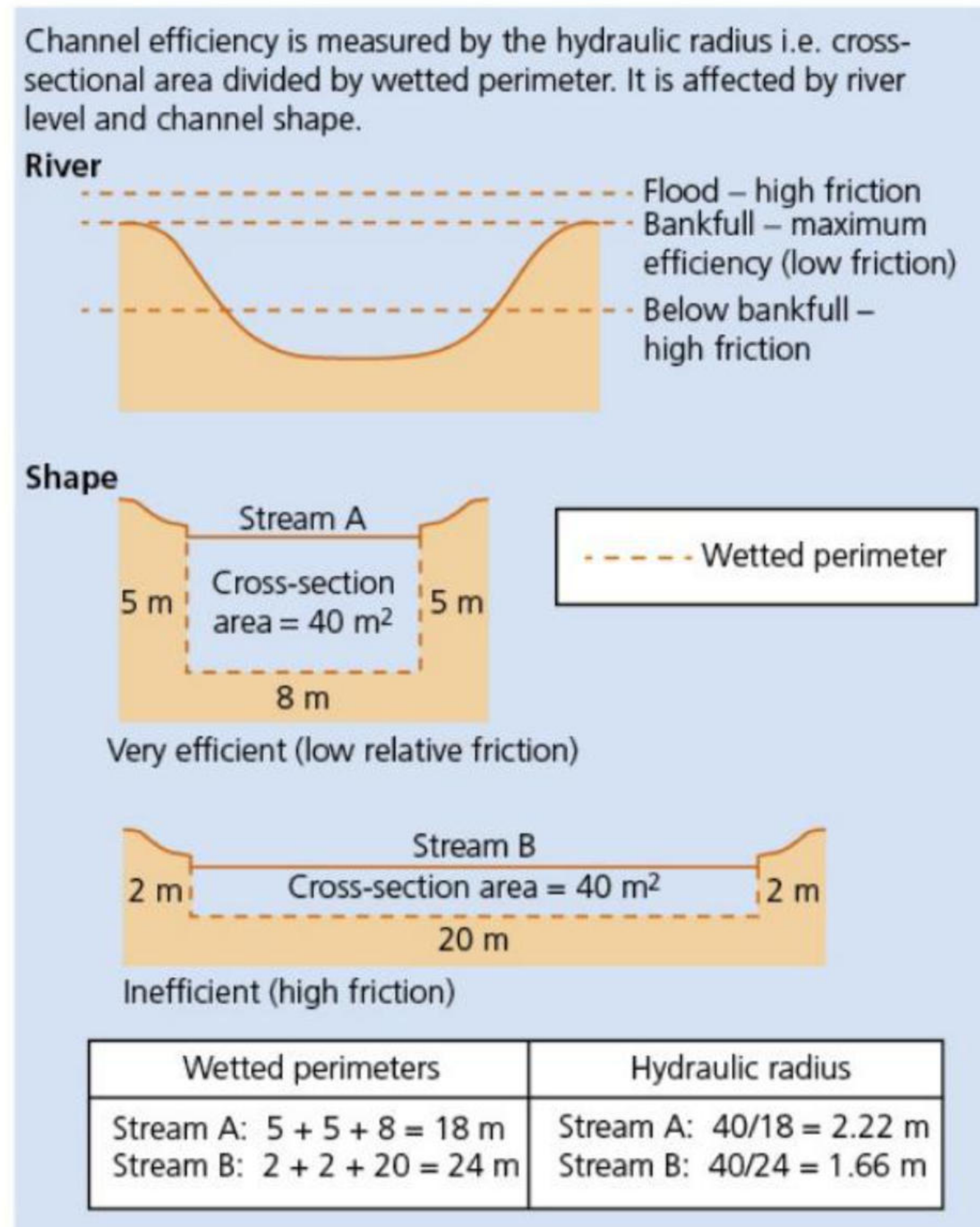
wetted perimeter. This results in less friction and reduced energy loss, and therefore greater velocity.

- » Stream B: With a smaller hydraulic radius, this stream has a larger amount of water in contact with the wetted perimeter. This results in greater friction and more energy loss, and therefore reduced velocity.

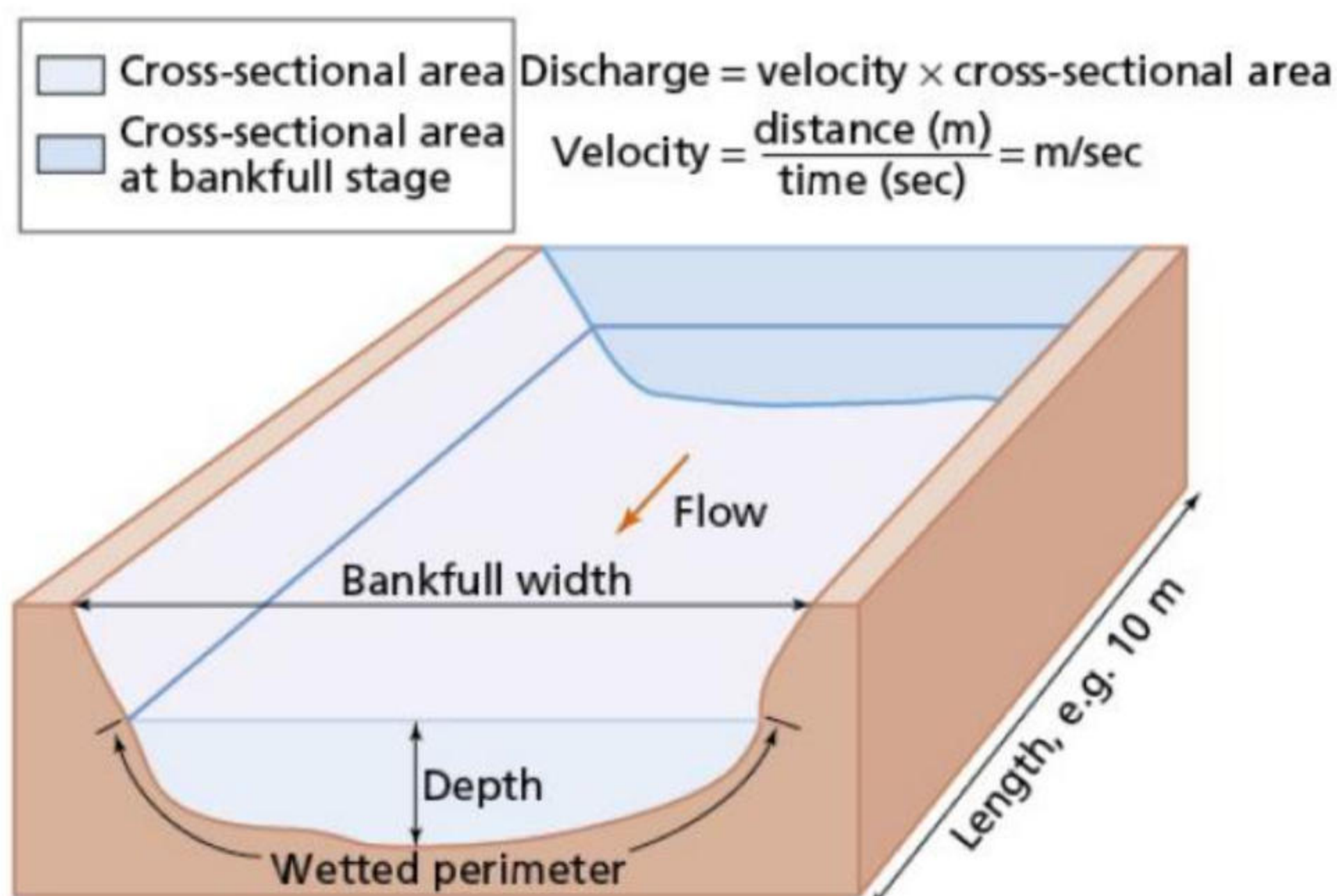
Stream A is more efficient than stream B.

There is a close relationship between the characteristics of the channel in which water is flowing and its **velocity** and **discharge**. These characteristics include the depth, width and channel roughness.

- » River velocity is the speed at which water is flowing, in metres per second (m/s).
- » Discharge ([Figure 1.7](#)) is the volume of water that passes through a section of the river per unit of time, expressed in cubic metres per second (m³/s). The discharge of a river is calculated by multiplying the river's cross-sectional area by its velocity.



▲ **Figure 1.6** The cross-sectional area and wetted perimeter



▲ **Figure 1.7** Discharge

Channel roughness

Channel roughness describes how rough (uneven) the bed of a river is. Channel roughness causes friction, which slows down the velocity of the river water. Friction is caused by irregularities in the river bed, such as boulders, pebbles, potholes and vegetation, and by contact between the water and the river bed and banks.

If a river bed was equally smooth from its source to its mouth, it would be reasonable to expect that the velocity of the river would be greater near the source because of the steeper gradients associated with upland source regions. However, in reality the reverse is true, due to the very high degree of channel roughness in upland areas compared to the relatively smooth channel beds usually found in the lowland sections of rivers.

Discharge normally increases downstream, as do width, depth and velocity. The increase in channel width downstream is normally greater than the increase in channel depth. Large rivers, with a higher width-to-depth (w/d) ratio, are more efficient than smaller rivers with a lower width-to-depth ratio, because less energy is spent in overcoming friction. Thus, the carrying capacity increases and a lower gradient is required to transport the load. Although river gradients decrease downstream, the load carried is smaller and therefore easier to transport.

Activities

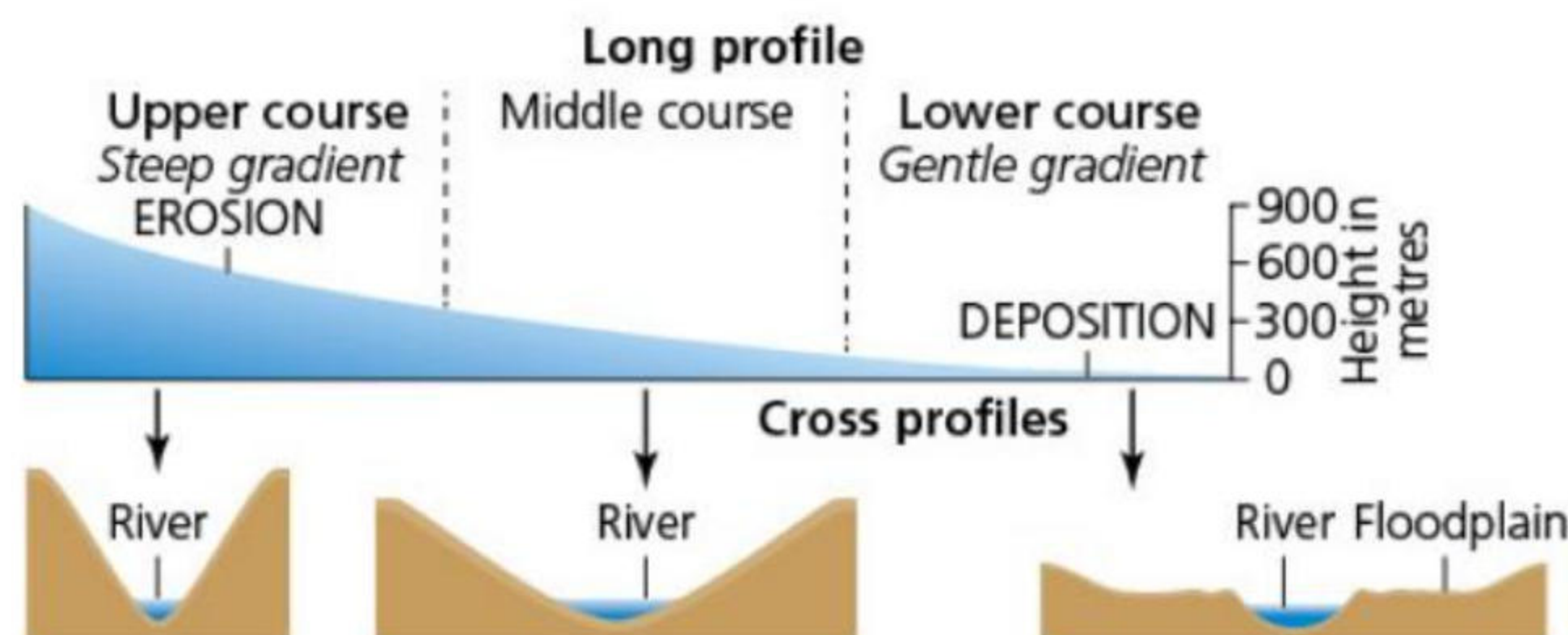
- 1 Describe three different sources of rivers.
 - 2 What is the long profile of a river?
 - 3 Explain the terms:
 - a Wetted perimeter
 - b Hydraulic radius.
 - 4 What is channel roughness and how does it affect the efficiency of a river?
-

The Bradshaw model

The **Bradshaw model** is a geographical model that suggests how a river's characteristics change from the source to the mouth of the river. [Figure 1.8](#) shows the generalised ways in which the long and cross profiles of a river change as the river's gradient decreases. The long profile is sub-divided into three sections:

- » Upper course
- » Middle course
- » Lower course.

The characteristics of each of these three sections are distinctly different, as are the physical processes operating in them. Look back at [Figure 1.5 \(page 4\)](#) to remind yourself of how factors such as width, depth, velocity and discharge change from upstream to downstream.



▲ **Figure 1.8** The long and cross profiles of a river

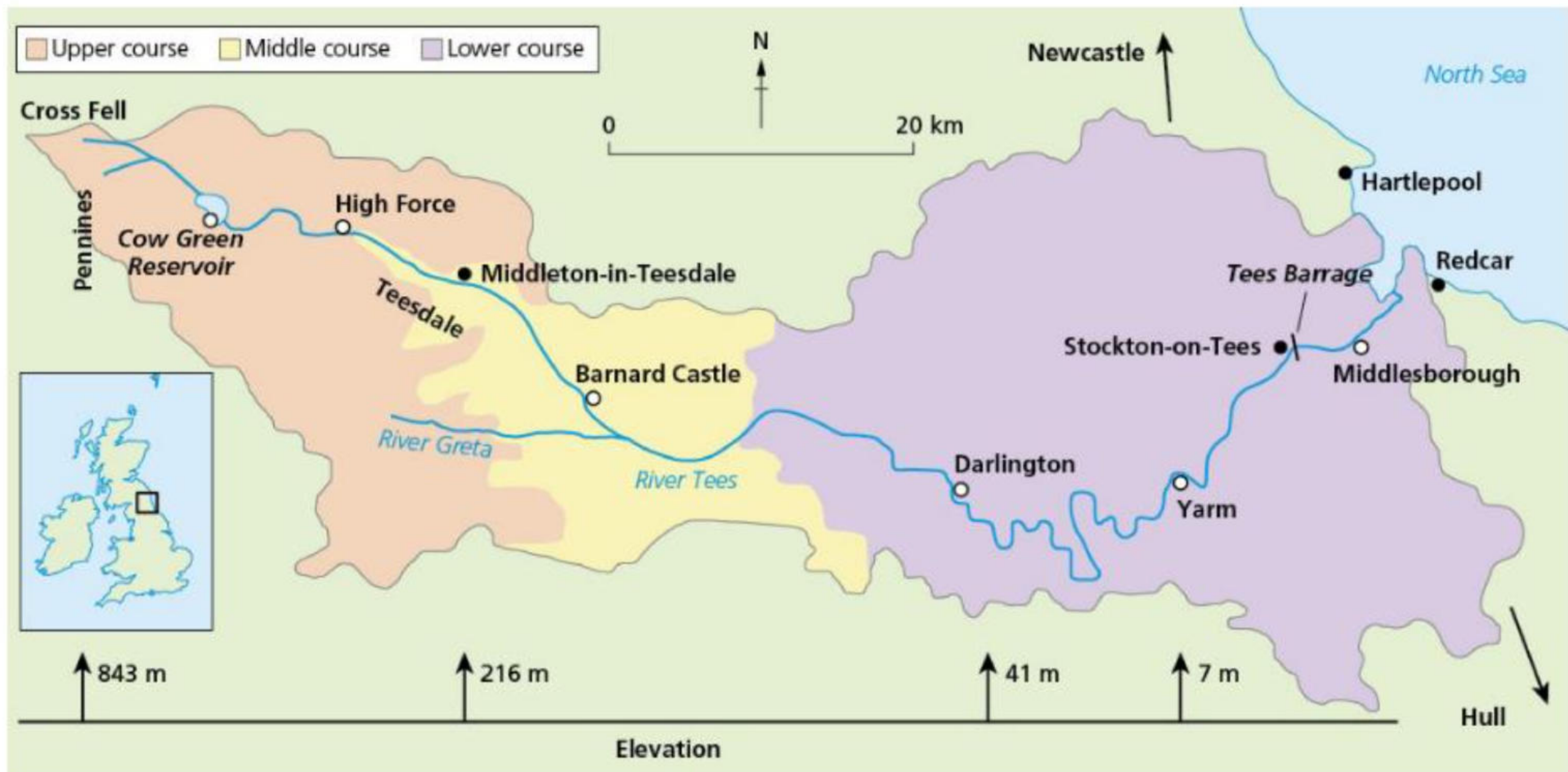
The River Tees is a reasonable exemplification of the Bradshaw model. The Tees is one of the major rivers in northeast England; it drains an area of about 1800 km². The source of the Tees is at Cross Fell, on the eastern side of the Pennine Mountains ([Figure 1.9](#)). The river rises at a height of over 750 m, flowing 160 km (channel length) eastwards to the mouth of the river in the North Sea. The Tees exhibits most of the classic processes and landforms of the upper, middle and lower courses of rivers.

The upper course

The upper course of the Tees is mainly an area of moorland, where the main land use is sheep farming. Annual precipitation can rise to over 2000 mm per year on the highest land. Precipitation decreases significantly eastwards towards the North Sea. The river channel is shallow and narrow. The bed is uneven, with sizeable angular boulders in places. There is much friction and the water flows more slowly here than further downstream, where the channel is wider, deeper and less uneven. High levels of friction upstream can cause considerable turbulence.

Vertical erosion has created a steep channel gradient and steep valley sides. These features, combined with impermeable rock, result in the river reacting quickly to rainfall. Impressive waterfalls are evident at Cauldron Snout and High Force, along with clear examples of interlocking spurs. High Force ([Figure 1.10](#)) is

the UK's largest waterfall, at 21 m high. A deep plunge pool has been eroded at the base of the waterfall. At High Force, a bed of hard rock (dolerite) overlies softer rock (sandstone and shale). Look at [Figure 1.21 \(page 13\)](#) to understand what happens when this occurs. As the waterfall has eroded upstream, it has left behind an impressive gorge downstream of High Force. Rapids are also in evidence in this section of the river.



▲ **Figure 1.9** Map of the River Tees from source to mouth



▲ **Figure 1.10** High Force waterfall, County Durham

The middle course

Below the town of Middleton-in-Teesdale, the valley widens out and the channel slope becomes more gentle. The fertile soils of the early stage floodplain provide for productive agriculture. Lateral erosion takes over from vertical erosion, forming distinctive meanders. Good examples can be seen near Barnard Castle. The Tees is joined by important tributaries including the Rivers Lune, Balder and Greta. The result is a substantial increase in the **volume** of water in the river.

The lower course

Here the channel gradient is gentle, with the river very close to sea level as it meanders across a fertile clay plain to its estuary between Hartlepool and Redcar, downstream of Middlesborough. The Tees is now predominantly tidal in nature. Deposition is the dominant process, evidenced by mud flats at low tide. The river has now formed much larger meanders across its wide floodplain, for example near Yarm. Oxbow lakes and levées are clearly evident. Just downstream of Yarm, the River Levin joins the Tees. The original winding river channel below Stockton has been straightened by artificial cuts to aid navigation. The mouth of the Tees is in the form of a large estuary with mudflats and sandbanks.

Activities

1 Study [Figure 1.9](#).

- a Describe the changes in gradient and altitude from the upper course to the lower course.
- b How does the cross profile of a river change from source to mouth?

2 Describe the source and the mouth of the River Tees.

How the drainage basin operates within the water cycle

Hydrology is the study of water. The Earth's water is constantly recycled in a **closed system** called the **hydrological cycle** (water cycle). A closed hydrological system means that the volume of water in the hydrosphere today is the same as has always been present in the Earth's atmosphere system.

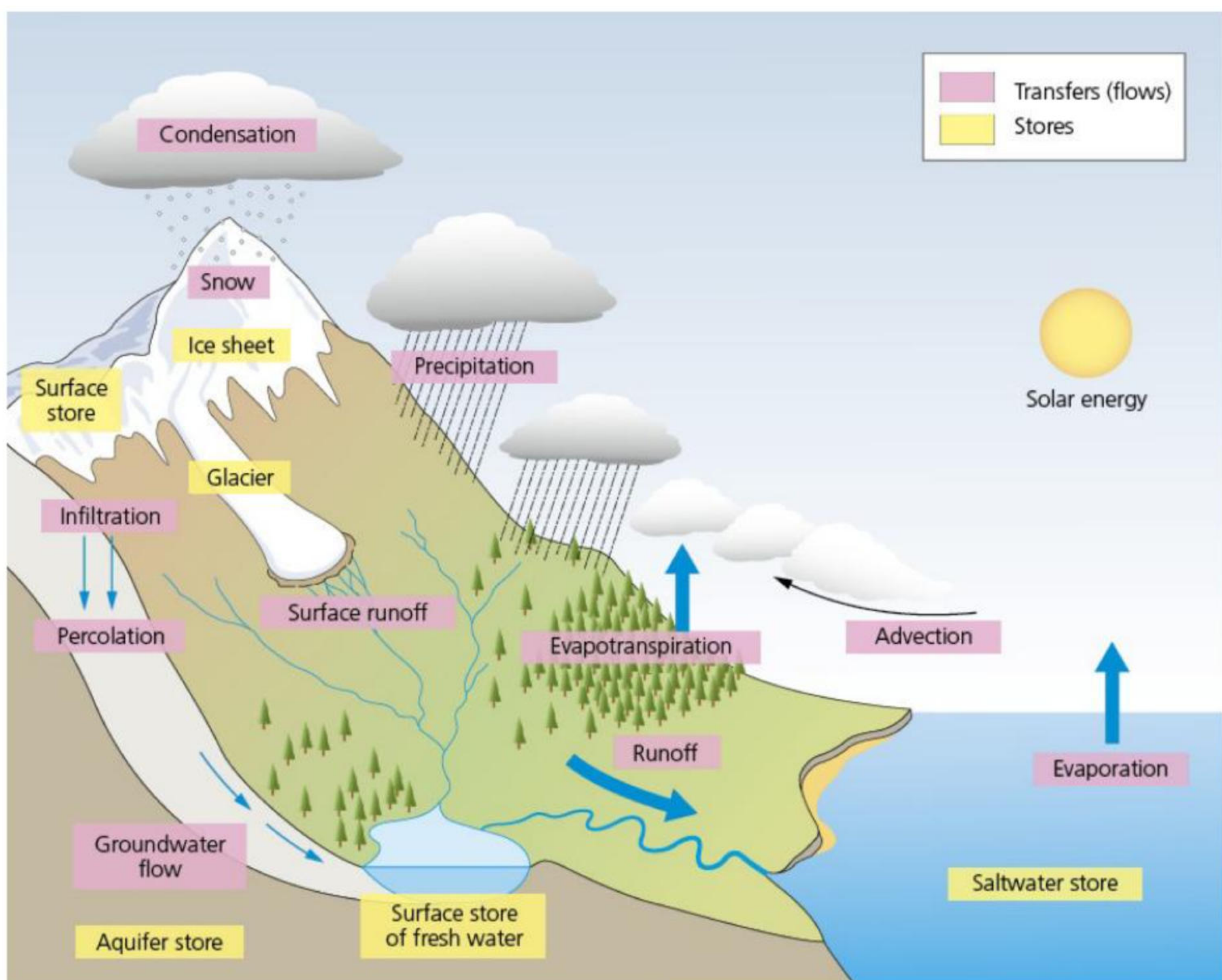
[Figure 1.11](#) shows that water can be held for varying periods of time in different **stores**, namely:

- » in oceans and seas
- » on land as rivers, lakes and reservoirs
- » in bedrock as groundwater
- » in the atmosphere as water vapour and clouds.

Over 97 per cent of the world's water is stored in oceans and seas. These water bodies make up about 70 per cent of the surface of the Earth. This water is, of course, saline. The rest of the world's water (less than 3 per cent) is fresh. Of this fresh water, more than 68 per cent is held as ice and snow, with most of this in Antarctica and Greenland, and just over 30 per cent as groundwater. Only about 0.3 per cent of fresh water is found in rivers, lakes and surface reservoirs.

Just 0.001 per cent of the world's water is held in the atmosphere at any one time. This amounts to only about 10 days' supply of average rainfall around the world. This means that without transfers in the hydrological cycle ([Figure 1.12](#)), the world would run short of fresh water very quickly.

- » Antarctica covers an area of almost 14 million km² and contains 30 million km³ of ice. This equates to around 61 per cent of all fresh water on Earth. The Antarctic ice sheet holds an amount of water such that if it were to melt, the sea level would rise by 70 m. The Greenland ice sheet covers 1.7 million km², which is about 70 per cent of the surface of Greenland.



▲ **Figure 1.11** Processes, stores and transfers in the hydrological cycle



▲ **Figure 1.12** Ice melting (a transfer) at the edge of Antarctica, by far the world's largest mass of ice

- » About 30 per cent of the Earth's fresh water is held as groundwater. At over 1.7 million km³, the Great Artesian Basin in Australia underlies 22 per cent of the country and is arguably the largest groundwater **aquifer** in the world.
- » Lake Baikal in eastern Russia is the largest-volume fresh water lake in the world. It is also the world's deepest lake. It covers an area of 31,500 km², with a maximum depth of 1637 m.

Transfers of water occur between stores by the following processes:

- » evaporation
- » condensation
- » precipitation
- » transpiration
- » overland flow
- » infiltration
- » percolation
- » throughflow
- » groundwater flow.

Evaporation, condensation and precipitation

These are the three main processes in the hydrological cycle. Water exists in three states – liquid, solid and vapour – and these three states are constantly interchanging.

- » **Evaporation** is the process by which liquid water is changed into a gas (water vapour). It takes place mainly from surface water. The energy required comes from the sun's heat and from wind. The higher the temperature, the greater the potential for evaporation. Look how quickly water evaporates from a concrete or tarmac surface on a very hot day compared with a cooler day. Evaporation is also faster on a windy day than on a calm day. Evaporation from water surfaces on land would not be enough to keep rivers and lakes full and to provide the human population with enough drinking water. Fortunately, large amounts of water evaporated from the seas and oceans are carried by air masses on to land, where condensation and precipitation take place.
- » **Condensation** is the process by which water vapour changes into water droplets. It happens when water vapour is cooled to a level known as the dew point. This is when clouds begin to form. The extent of cloud cover at any point in time is a good indication of the intensity of condensation in the atmosphere. Clouds are tiny water droplets suspended in air, while rain droplets (precipitation) are much larger. This larger size enables rain droplets to overcome rising currents in the air in order to reach the ground surface.
- » **Precipitation** occurs when water in any form falls from the atmosphere to the surface. This is mainly as rain, snow, sleet and hail. Thus, water is constantly recycled between the sea, the atmosphere and the land. The main characteristics that affect local hydrology are the amount of precipitation, seasonality, intensity, type (for example snow or rain) and variability.

The processes that operate in a drainage basin

While the global hydrological system is a closed system, the hydrological cycle of an individual drainage basin is an open system as it is open to external inputs and outputs. The system has a range of:

- » inputs – water entering the system
- » stores – places where water is held in the system

- » transfers (flows) – where water is flowing through the drainage basin system
- » outputs – where water is lost to the system.

Precipitation is the input to the system. When precipitation reaches the surface, it can follow different pathways:

- » A small amount falls directly into rivers as **direct channel precipitation**. This adds to **channel flow**, which is the movement of water within the river channel.
- » The rest falls on to vegetation or the ground.



▲ **Figure 1.13** A waterlogged field, near the town of Navan, Republic of Ireland – the soil in this field was already saturated when heavy rain fell in the night before the photograph was taken

If heavy rain has fallen previously and all the air pockets in the soil are full of water, the soil is said to be **saturated** (Figure 1.13). Because the soil is unable to take in any more water, the rain flows on the surface under the influence of gravity or remains on a flat surface in a waterlogged state. Flowing surface water is called **surface runoff** or overland flow.

If the soil is not saturated, rainwater will soak into it through the process of **infiltration**. It then moves vertically down through the soil and rock by the process of **percolation**. If the rock below the soil is **permeable** (meaning it allows water into it), the rainwater will continue to soak down deeper into the rock. This water will eventually come to **impermeable** rock (which does not allow water into it). The underground water level will build up towards the surface from here. This water does not remain stationary but flows downslope under gravity. The upper level of underground water is the **water table**. Water contained in rocks is known as **groundwater**, and water on the move in rocks is called groundwater flow. Rock that holds groundwater is known as an aquifer.

A spring occurs when underground water emerges at the surface.

- » This happens where a permeable rock such as limestone covers an impermeable rock such as clay. Rainwater that can percolate into the permeable rock is unable to penetrate the impermeable rock below. This water will emerge at the surface as a spring, provided the water table is above the surface level.
- » This also happens when the water table in a normally dry area reaches the surface during a period of unusually heavy rain. Such springs generally flow for only a short period of time.

Rainwater can be intercepted by vegetation. **Interception** is greatest in summer, when trees and plants have greater leaf coverage.

- » Some rainwater will be stored on leaves and then evaporated directly into the atmosphere.
- » The remaining intercepted water will either drip to the ground from leaves and branches or it will trickle down tree trunks or plant stems (**stemflow**) to reach the ground.

Vegetation takes in moisture through its root system. It loses some of this into the air by **transpiration**. Surface water is also lost by evaporation. The combination of these two processes is known as **evapotranspiration**.

In some countries precipitation is fairly regular throughout the year. However, in other countries there may be distinct wet and dry seasons. In these areas, rivers may dry up completely for many months. In **deserts**, small river channels may be dry for most of the year.

Activities

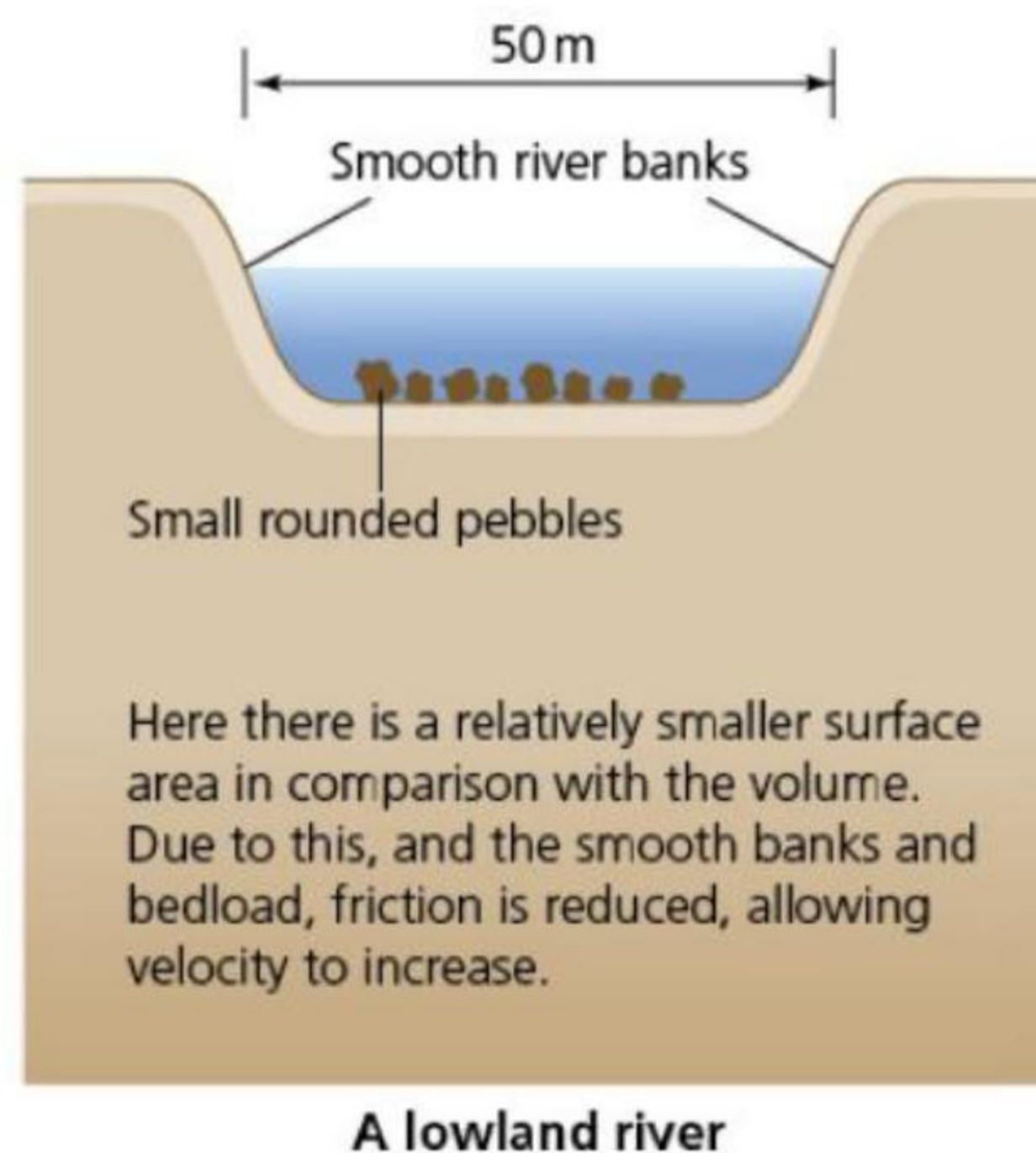
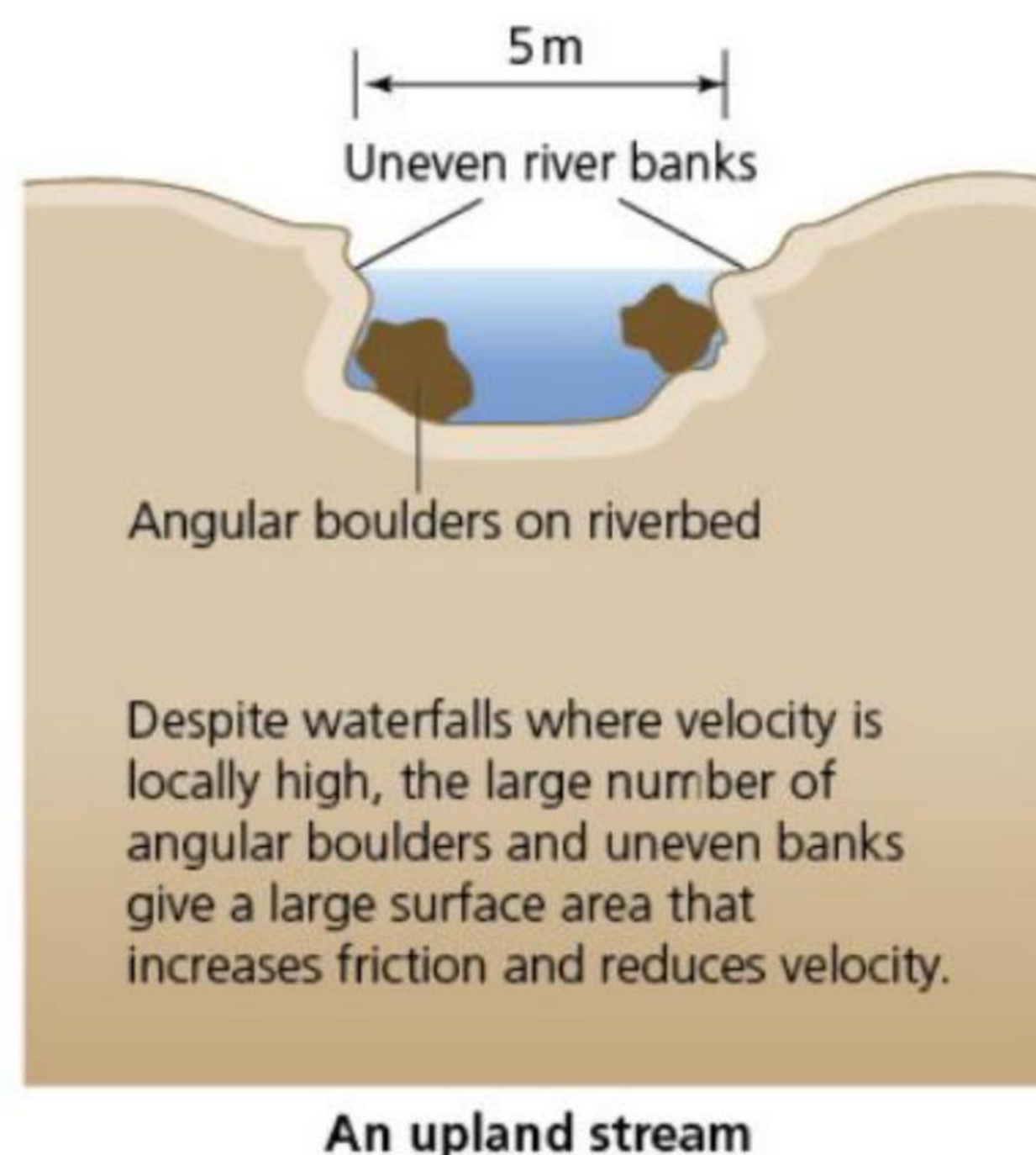
- 1 Study [Figure 1.11 \(page 7\)](#).
 - a List three stores and three transfers (flows) in the hydrological cycle.
 - b Draw a labelled diagram to show the relationship between evaporation, condensation and precipitation.
 - 2 Why is the global hydrological system a closed system, while the hydrological cycle of an individual drainage basin is an open system?
 - 3 Explain the differences between:
 - a overland flow and groundwater flow
 - b infiltration and percolation.
 - 4 What is evapotranspiration?
-

The processes that operate within a river

Energy is needed for transfers to occur. Around 95 per cent of a river's energy is used to overcome **friction**. The remaining 5 per cent or so is used to erode the river channel and transport this material downstream. The amount of energy in a river is determined by:

- » the amount of water in the river
- » the speed at which it is flowing.

[Figure 1.14](#) shows that in the upper course of a river, near the source, a river's channel is shallow and narrow, and the river bed is often strewn with boulders and very uneven. There is a lot of friction so the water flows more slowly here than it does further downstream in the middle course and lower course, where the channel is wider, deeper and less uneven. [Figure 1.15](#) shows a river in its upper course – notice the steep gradient, the boulders in the river and the amount of 'white water'. The latter is a good indication of a high level of friction.



▲ **Figure 1.14** Velocity and discharge in the upper and lower courses of a river



▲ **Figure 1.15** A river in its upper course – British Columbia, Canada

Erosion

There are four processes of **erosion** that take place in a river:

- » **Hydraulic action**: The sheer force of the river water removing loose material from the bed and banks of the river.
- » **Abrasion/corrasion**: The wearing away of the river bed and banks as a result of the river's load hitting them repeatedly.
- » **Attrition**: In swirling water, rocks and stones collide with each other and with the bed and banks. Over time, the sharp edges become smooth and the rocks and stones become smaller in size ([Figure 1.16](#)).
- » **Solution**: Some rocks, such as limestone, which is soluble in slightly acidic water, dissolve slowly in river water.



▲ **Figure 1.16** Rocks in a dry section of river bed that have been rounded and reduced in size by attrition – Kyrgyzstan, Central Asia

Along the course of a river there are two main types of erosion that take place:

- » **Vertical erosion** (downward): This takes place in the upper course of the river near the source, where the river cuts down into its bed and deepens the valley.
- » **Lateral erosion** (sideward): This takes place in the middle and lower courses and widens the valley.

Most erosion occurs when discharge is high and rivers are in **flood**.

Transportation

There are four processes by which a river can transport its **load** ([Figure 1.17](#)):

- » **Solution**: In areas of calcareous rock (limestone), material is carried in solution as dissolved load.

- » **Suspension**: The smallest particles (silt and clay) are carried in suspension by the moving water.
- » **Saltation**: Larger particles (sand, gravel, very small stones) are transported in a series of 'hops' or bounces.
- » **Traction**: Large stones are pushed along the bed by the process of traction.

Parts of the load that are moved by traction when the discharge of the river is low may be transported by saltation when the discharge is high.

Deposition

Deposition takes place when a river does not have enough energy to carry its load. This can happen when:

- » the gradient decreases
- » the discharge falls during a dry period
- » the current slows down on the inside of a meander
- » the river enters a lake or the sea.

When a river loses energy, the large, heavy material known as the **bedload** is deposited first. Lighter material is carried further downstream. The gravel, sand and silt that are deposited is called **alluvium**. This is spread over the floodplain. The load transported by solution is carried out to sea with much of the clay and the lightest suspended particles.

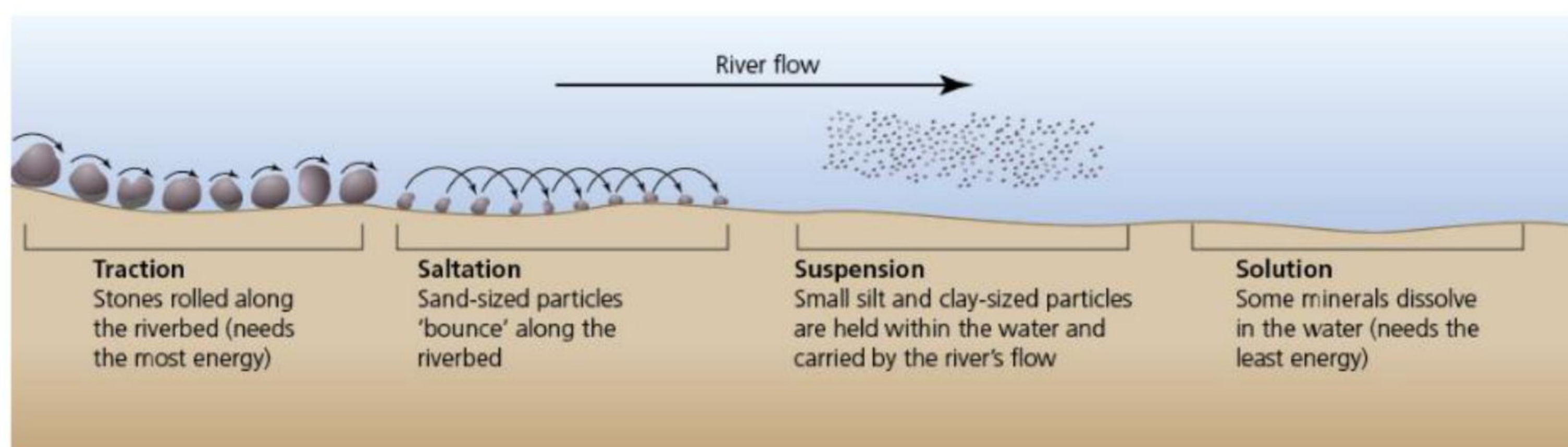
[Table 1.2](#) gives examples of the factors affecting the processes of erosion, **transportation** and deposition.

▼ **Table 1.2** Examples of factors affecting processes

Factor	Effect on erosion, transportation and deposition
Climate	● Heavy rainfall leads to higher discharge, which leads to increased action of river processes. ● Higher temperature leads to increased evaporation, which leads to lower discharge and reduced action of river processes.
Slope	● Steep slopes result in fast-flowing rivers with strong erosive power. ● Gentle slopes encourage deposition.
Geology	● Rivers erode valleys made of soft rock at a rapid rate. ● Very porous (e.g. chalk) and permeable (e.g. carboniferous limestone) rocks may lack surface river flow for all or part of the year.
Altitude	● Snowmelt and melting glaciers have a big impact on river regimes and processes.
Aspect	● South-facing slopes (in the Northern Hemisphere) have higher rates of evaporation and transpiration, which can affect discharge.

Activities

- 1 a Define friction.
b How much of a river's energy is used to overcome friction?
- 2 Describe and explain the differences between the two diagrams in [Figure 1.14 \(page 10\)](#).
- 3 List the four processes of:
 - a erosion
 - b transportation.
- 4 Under what conditions is deposition likely to occur?



▲ **Figure 1.17** The processes of transportation